

Software Project Management – IS212

Introduction

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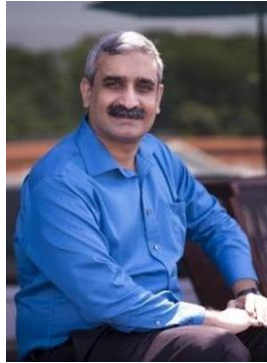


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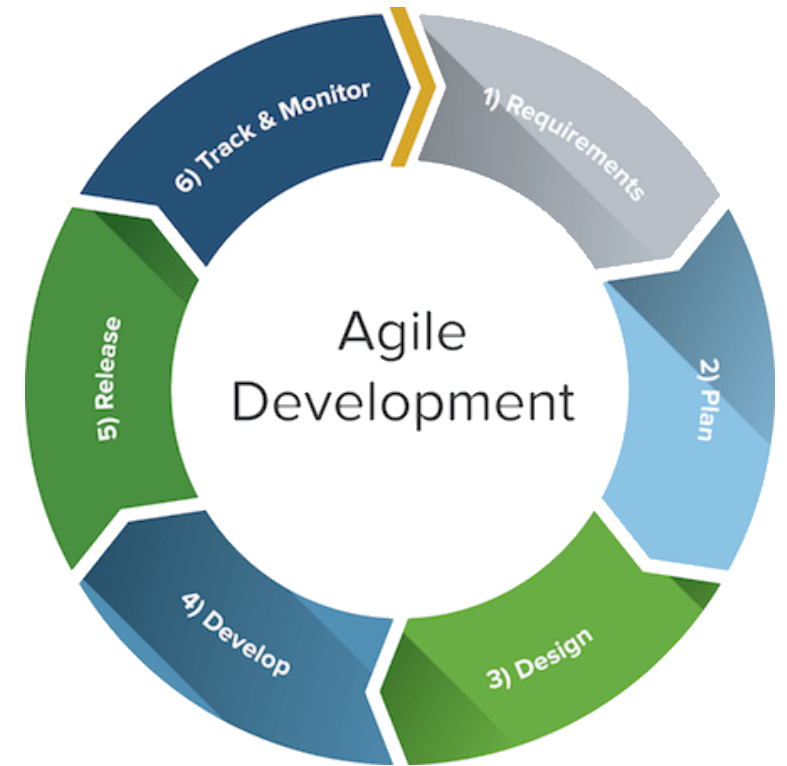


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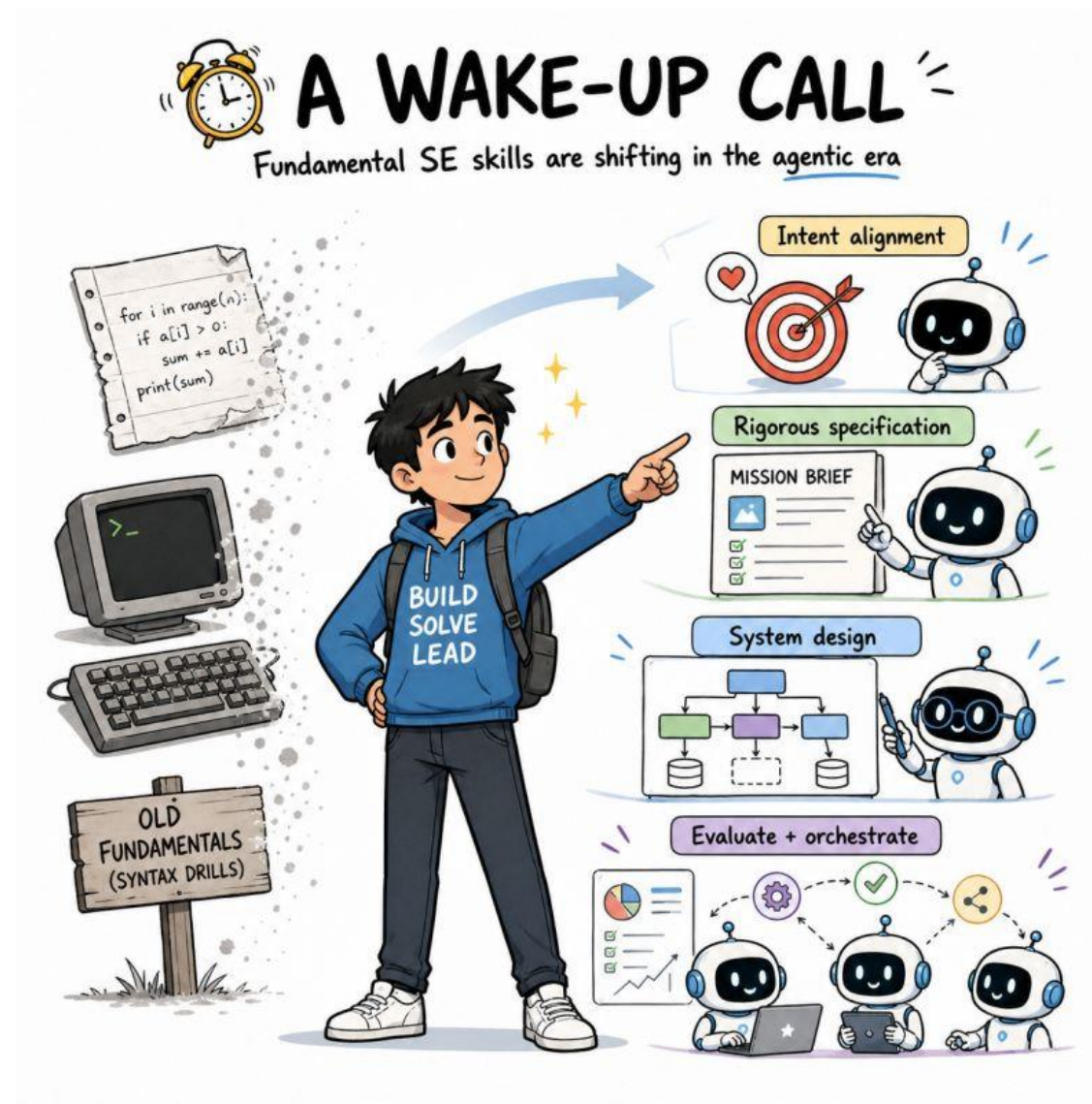
Learning Objectives

- **SPM in a nutshell:** hands-on *agile* software engineering and project management
- Learning objectives:
 - Apply a range of **agile methods** to develop and manage a software project
 - Manage **complexity and uncertainty** in projects using the **scrum** framework
 - **Design and document** software systems using appropriate notations and abstractions
 - **Collaborate** with product owners, scrum masters, and other developers to deliver value
 - Ensure software quality through **testing, continuous integration and delivery**, and the **review and verification of (AI-assisted) code**



<https://www.smartsheet.com/agile-vs-scrum-vs-waterfall-vs-kanban>

Software Engineering is Changing



traditional vs. agile methods

managing “WIP”

user stories

*reviewing
AI-assisted code*

modelling with C4 and UML

Core Competencies

Git & GitHub Actions

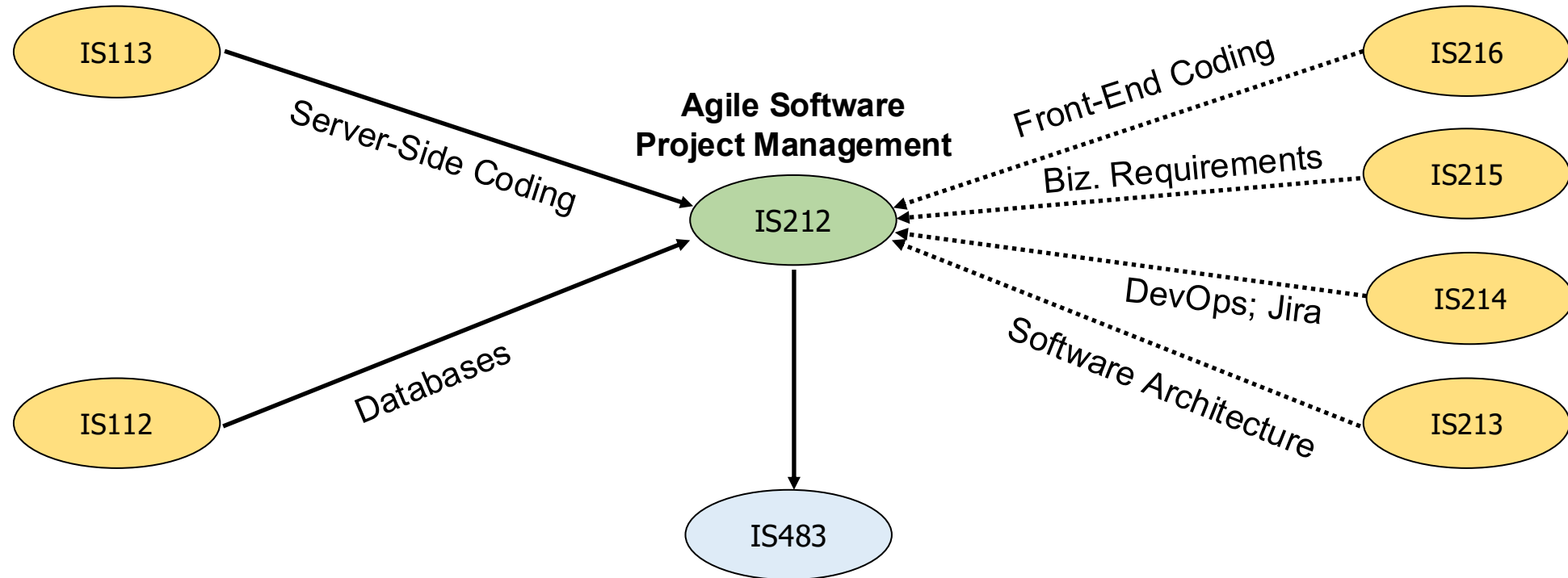
*estimation &
planning poker*

test-driven development

continuous integration & delivery

code smells & refactoring

Where IS212 Fits



Schedule

Week	Topic	Assessment Milestones
1	Software Development Processes	<i>Customer briefing released (Fri 6pm)</i>
2	User Stories	<i>Individual assignment #1</i>
3	Agile Estimation; Scrum Meetings	<i>Mini-Quiz #1; Project instructions released (Fri 6pm)</i>
4	Test Cases; Software Architecture	
5	UML Diagrams; Collaborating with Git	<i>Mini-Quiz #2</i>
6	Automated Testing #1; Continuous Integration (CI)	<i>Individual Assignment #2</i>
7	Scrum Project Consultations	

Schedule

Week	Topic	Assessment Milestones
9	Automated Testing #2; Continuous Delivery (CD)	<i>Mini-Quiz #3</i>
10	Code Review & Technical Debt (Agentic Era)	<i>Topic Research & Sharing</i>
11	Code Refactoring	<i>Mini-Quiz #4</i>
12	– <i>no class</i> –	<i>Project Deliverables Due</i>
13	Closed Door Q&A with Instructors	<i>Project Q&A Session</i>
14	– <i>study week</i> –	<i>Intra-team peer evaluation</i>
15–16	Exam Period	<i>Final Exam</i>

Labs – Software and Tools

- The course concepts will be demonstrated through a mix of **hands-on exercises** and **labs**
- We will use the **Python** language and **Flask** library for our examples
- You are welcome to explore **different software stacks** tools in the project



Our Policy on AI Tools

- You are **welcome to use generative AI tools** in this course except in proctored assessments (e.g. quizzes, exam, project Q&A)
 - Refer to assessment briefings on how AI usage must be declared
- SMU has an institutional subscription to **Google Gemini**  **Gemini**
 - <https://gemini.google.com/>
 - Log in using your SMU email address without the school sub-domain
 - E.g. “madonna.1976@computing.smu.edu.sg” → “madonna.1976@smu.edu.sg”
- It's still important to **learn the fundamentals** so that you can ‘verify’ that **Gemini**, **Claude**, etc. are actually generating the right thing 😊

AI Coding Assistants (as of July 2026)

Tool	Runs in	What you get for free	Paid	Watch out for
GitHub Copilot	VS Code, Visual Studio, JetBrains, CLI	Copilot Student — free while your GitHub student status is verified	\$10	~200 AI credits/month; no model choice (“Auto” only)
Windsurf	Own editor; VS Code / JetBrains plugins	Free tier, no expiry: unlimited tab completions, light agent use	\$20	Cascade (the agent) runs out fast — tab completions don’t
Google Antigravity	Own editor (VS Code-based) + CLI	Free, no card needed	\$20	Free quota cut from 250 to ~20 agent requests/day. Don’t rely on it
OpenAI Codex	VS Code extension, CLI, web	Included in ChatGPT Free — enough to try, not to depend on	\$20	Billed by tokens; limits reset on a rolling 5-hour window
Cursor	Own editor (VS Code-based)	Hobby tier — limited agent requests, quota not published	\$20	Free year for students closed to new sign-ups (25 Jun 2026)
Claude Code	Terminal; VS Code / JetBrains extension	Nothing — no free tier	\$20	Strongest on large, multi-file work, but you must pay to use it

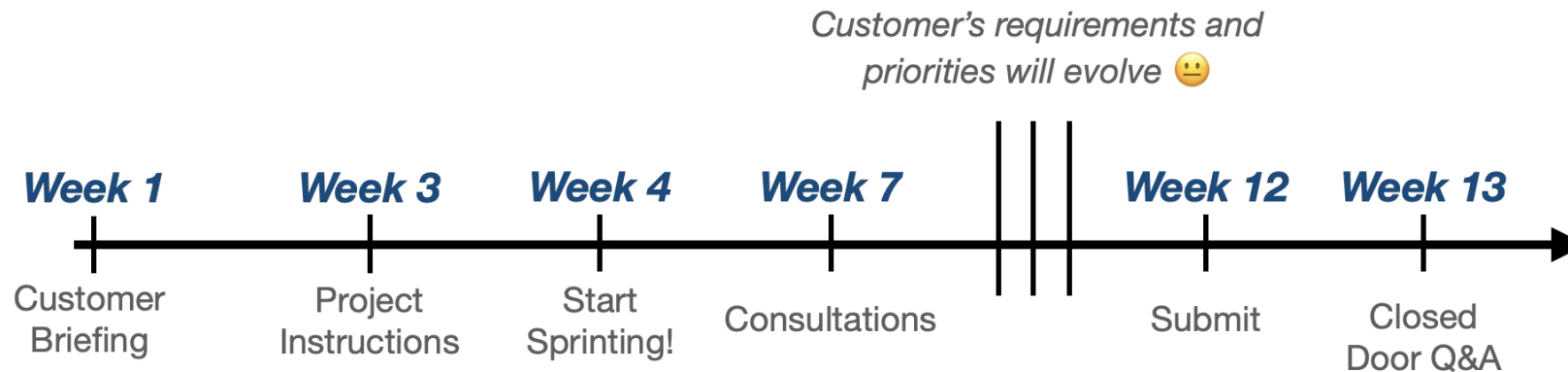
US\$/month, cheapest individual paid plan. Prices and free limits change constantly — check before you commit.
 NB: SMU’s Gemini subscription only covers the chat interface, not IDE-based coding assistants.

Assessments – Scrum Project

- Significant group project (35%)
 - Form own teams (5–6 pax); maximum 8 groups per section
 - Designed to simulate a real-world scrum project – the good, the bad, the ugly
- **Week 1** – customer briefing released (Friday 6pm)
 - **Warning!** This will initially be **deliberately ambiguous**
 - The ‘customer’ (one of your instructors) will schedule consultation sessions to answer your questions
 - ‘Customer’ clarifications may differ across sections – **only the clarifications of your section matter**
 - Refine your understanding of the requirements but do not start coding yet!
- **Week 3** – project instructions released (Friday 6pm)
 - Specifies the scope of the system expected for the first release (i.e., Week 12)...
 - ...and the scrum artefacts expected in your final submission
 -  The customer **will make changes to some requirements** in subsequent weeks
 - **IS212 emphasises process**: a great system built with an ad hoc process will **not** score well

Assessments – Scrum Project

- **Week 4** – start sprinting! **Four** 2-week sprints (or similar) will be enough (must finish by Friday Week 12)
- **Week 7** – **informal consultations** on your scrum process (i.e. feedback on first sprint)
- **Week 12** – all project deliverables due (Fri 23:59), including recordings of the meetings from your final sprint
- **Week 13** – closed-door Q&A sessions with your instructors. We'll ask about your team's process / outcomes 😊



Assessments – Individual Assignments & Quizzes

- Individual Assignments (5%)
 - **Week 2:** a small number of initial “user stories” (agile artefacts for capturing software requirements) based on the customer briefing
 - **Week 6:** exercise on writing automated tests and achieving code coverage
- Quizzes (12%)
 - Four short mini-quizzes conducted over the term (Weeks 3, 5, 9, 11)
 - Conducted at the start of lessons using eLearn and Respondus LockDown Browser
 - Only your best three quiz results will count
 - Quizzes can only be taken physically in your section’s class; no make-ups; no extra time when late
 - **Important:** sharing the quiz password to students outside the class gets a **score of 0 for both students**
 - **What if I can’t make a quiz?**
 - First absence (*any reason*) – quiz dropped from overall calculation (only the top 3 quizzes count)
 - Subsequent absences (*valid reason*) – quiz mark pro-rated using Final Exam Multi-Choice Questions
 - Subsequent absences (*invalid reason*) – **score of 0** for the quiz

Assessments – Topic Research & Sharing

- Topic Research & Sharing (3%)
 - **Week 10:** short group presentation on something recent or emerging in AI-assisted software development
 - E.g. a new tool, workflow, agentic system, or technique
 - Teams will explain what it is, demonstrate it (where possible), and share why it's interesting and how it fits into modern software delivery
 - Chance to learn from each other in this fast-moving field!

Assessments – Exam

- Final exam (35%)
 - 2hr closed book exam
 - Multi-Choice Questions, Written Questions, and a Case Study
 - Covers all materials and labs
 - We will provide sample questions for practice
- Tentative date: Thursday 26th November (1pm)
 - Refer to OASIS for your confirmed date and venue

Assessments – Class Participation

- Class participation (**10%**)
 - Attendance (compulsory in all weeks, except for valid reasons)
 - Active participation during classes and labs; peer learning (e.g. **Slack**); submission of **weekly reflections**
 - Please **remember your name tents!**
 - We value **quality of contributions over quantity**
 - Grading based on our holistic impression (not just bean counting)

	A+ / A	A- / B+	B / B-	C+ and below
Overall Impression	Significantly advances learning of the class.	Somewhat advances learning of the class.	Little contribution to learning of the class.	No contributions and/or invalid absences.
Quality of Interactions	Creative, insightful, stimulating, novel.	Relevant, interesting, thoughtful.	Irrelevant.	None to recall.

“I” grades

- No make-ups are possible for any assessments
- If you are unable to take the final exam for a valid, documented reason, you can apply for an “I” grade
- This is subject to the examination policies / approval of SMU

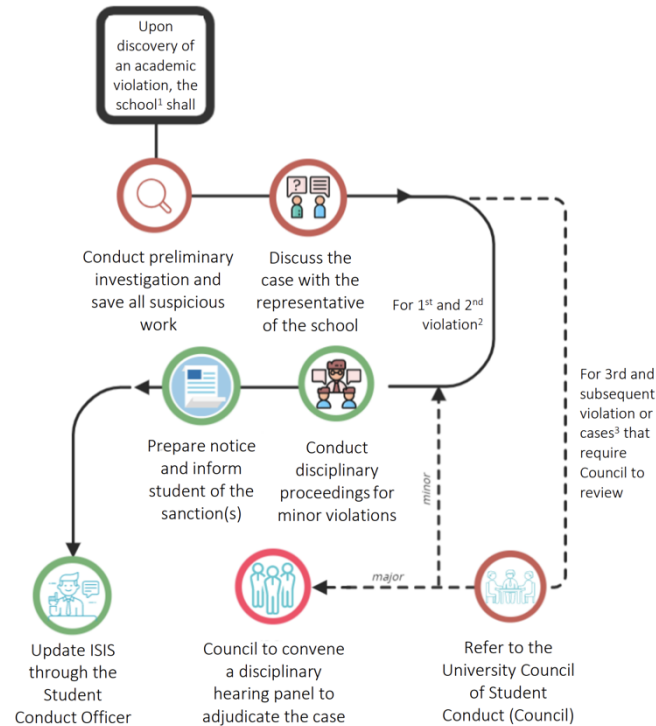
Support – Slack

- All nine sections will use Slack
- Fastest way to get human support
- Learn by answering questions and reading discussions
- Share interesting articles, news, Reddit discussions etc. in the **#interesting-links** channel
- **TO DO:** please upload a photo and add your section number (e.g. “(G2)”) after your name



<https://is212-2026.slack.com/>

REPORTING PROCEDURE



¹ The school who offers the module shall conduct the investigations and consult the Associate Dean of the school where the student is from when deciding on the sanction.

² For academic cheating, a 2nd violation shall be referred to the Council for classification.

³ The school may refer cases of 1st or 2nd violation to the Council should there be other factors that warrants more severe sanctions e.g. severity of the infraction, the nature and extent of the harm caused.

- [SMU Student Policies and Guidelines](#), [SMU Code of Conduct](#), [SMU Code of Academic Integrity](#)

Students should comply with the above policies; any violation will be reported. For academic violation reporting, see graphic on the left.

- **Punctuality**
All individuals are expected to be punctual. A delay of **15 minutes** or more will be considered as an **absence**.
- **Class Participation**
Class participation is an important aspect of the university's assessment system. Students are expected to participate actively in class.
- **Absence Policy**
Three or more unexcused absences may result in a failing grade. Valid excuses include illness, family emergencies, or other circumstances deemed acceptable by the University. See [Attendance Policy](#).
- **Communication Policy**
It is imperative to maintain a communication environment free from toxic content. This includes but is not limited to any form of communication that may be considered harmful, disrespectful, or inappropriate. Such content is not tolerated in any of our communication channels, and its presence may result in disciplinary action.

- **Bullying, Harassment And Sexual Misconduct**

SMU endeavors to provide a safe study and work environment for all members of our community. See <https://voices.smu.edu.sg/>

- **Accessibility**

If you anticipate or experience physical or academic barriers due to disability, please contact the university's disability services team if you have questions or concerns about academic provisions: DSS@smu.edu.sg. See also <https://smu.sharepoint.com/sites/oasis/SitePages/DSS/Student-Support.aspx>
 Please be aware that the accessible tables in our seminar room should remain available for students who require them.

- **Mrs Wong Kwok Leong Student Wellness Centre**

The Student Wellness Centre is dedicated to providing students with top quality mental health support. This facility will provide safe spaces for students to explore and manage their personal lives through attending sessions with professional counsellors or simply to have a mental repose from the daily rigours of life. See <https://studentwellness.smu.edu.sg/>

- **Student Services Hub**

The Student Services Hub (SSH), located at Basement 1 Concourse of SMU Li Ka Shing Library, serves as the University's one-stop resource, knowledge and student services centre. At SSH, you will be able to seek assistance on just about any student-related issue. See <https://studentservices.smu.edu.sg/>

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The selling of these materials and/or any copies thereof are strictly prohibited under Singapore copyright laws. Printed materials and electronic materials are both protected by copyright laws. All students are subject to Singapore copyright laws and must strictly adhere to SMU's procedures and requirements relating to copyright.

Please also note that for some materials, the publishers may specifically state that each copy is for the personal use of one individual only and no further reprographic reproduction is allowed, including for personal use. These restrictions are spelt out clearly on these specific sets of resources and students are required to adhere to these rules.

Students who infringe any of the aforesaid rules, laws and requirements shall be liable to disciplinary action by SMU. In addition, such students may also leave themselves open to suits by copyright owners who are entitled to take legal action against persons who infringe their copyright.

We strongly urge all students to respect the copyright laws and abide by SMU's procedures and requirements relating to copyright.